

AIAA 2205: Introduction to AI

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Assistant Professor



Enyan Dai

Trustworthy AI

Graph Mining

AI4Social Good

AI4Science

- ❑ Ph.D. from the Pennsylvania State University
- ❑ Published over 30 papers including **PAKDD Best Paper, Oral in ICLR, KDD, NeurIPS, WWW, TKDE** etc.
- ❑ Google Scholar Citation: ~2000, H-index: 18

Hobbies:



Football

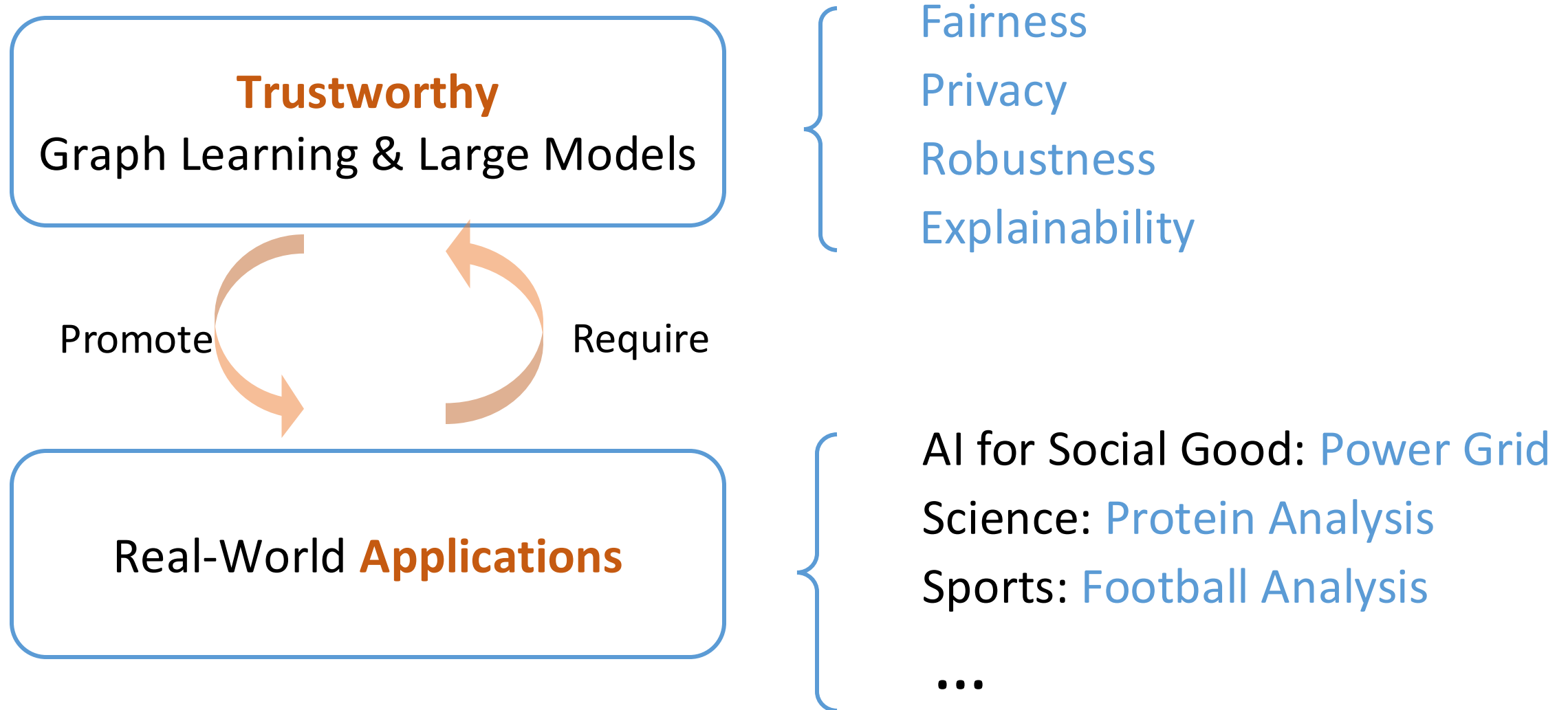
North London is Red!



Texas Holdem Poker
All in!

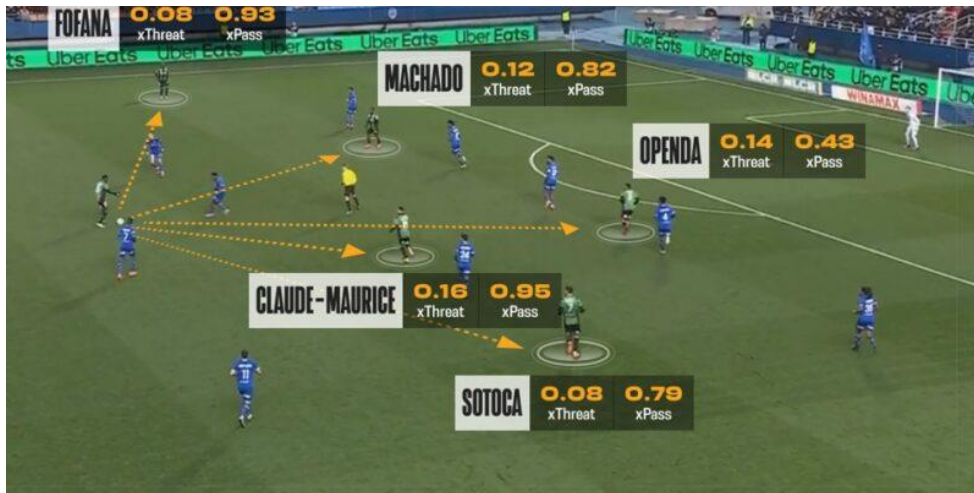
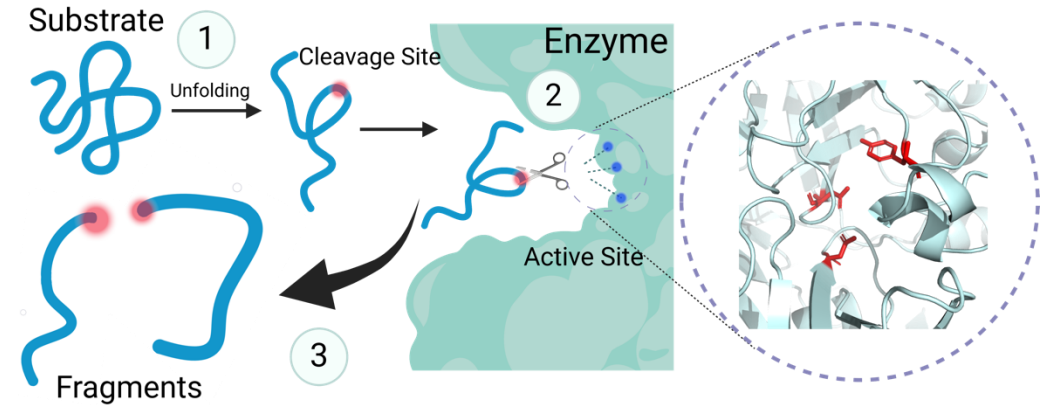
They all need trustworthy AI 🗣️

Research Directions at TAI (Trust & Application AI Lab)



We are Hiring

- Explainable Protein Modeling



- Line Breaking Passes Suggestion
- Goal Trending Estimation
- Tactics Suggestion

Outline of this course

1. Info about This Course

2. Introduction to Artificial Intelligence (AI)

Intending Learning Outcomes

1. Demonstrate a comprehension of advanced knowledge of Artificial Intelligence.
2. Demonstrate a comprehension of applications of Artificial Intelligence.
3. Recognize the limitations of current methods of Artificial Intelligence.
4. Apply programming and Artificial Intelligence skills.
5. Develop a broad interest in Artificial Intelligence and connect this knowledge to their major field of study.
6. Communicate effectively in written format to convey scientific knowledge and the application of modern technologies.

Grading Policies

- **Individual Assignment 1 (30%)**
- **Individual Assignment 2 (30%)**
- **Final Group Project (40%)**
 - Up to 5 members
 - Hand in project report
 - Require to do the final project presentation
- **Late and Resubmission Policy: Not permitted**

Policies

- Usage of **LLMs and agents** to help your learning process / homework / projects: 使用LLMs和智能体辅助学习过程/作业/论文:
 - **NOT forbidden** 不禁止; **Encourage** 鼓励
 - Required: Declaration of the usage 要求: 声明使用情况
 - Show good understanding and skills in using LLMs and agents 展示对LLMs和智能体的良好理解和使用技能
 - Clear and comprehensive 清晰且全面
 - Grading of the homework/projects will be graded based on 作业/论文的评分将基于:
 - The homework/projects itself 作业/论文本身
 - The declaration part 声明部分
- Interruption and questions in class 课上打断和提问
 - Welcome 欢迎
- Cheating and plagiarism 作弊与欺诈
 - Zero tolerance 零容忍
 - *e.g.*, (not limited to): Homework, data or False declaration of LLM usage

Office Hours

Instructor

Name	Email	Office	Office Hours
Enyan Dai	enyandai@hkust-gz.edu.cn	E4-407	Tuesday 3:30 PM -4:30PM

Wechat Group of TAs

群聊: AIAA2205

Teaching Assistant

Name	Email	Office Hours
Linxiao Cao	lcao950@connect.hkust-gz.edu.cn	Wednesday 6PM-7PM
Jiahua Lu	jiahualu@hkust-gz.edu.cn	Tuesday 6PM-7PM

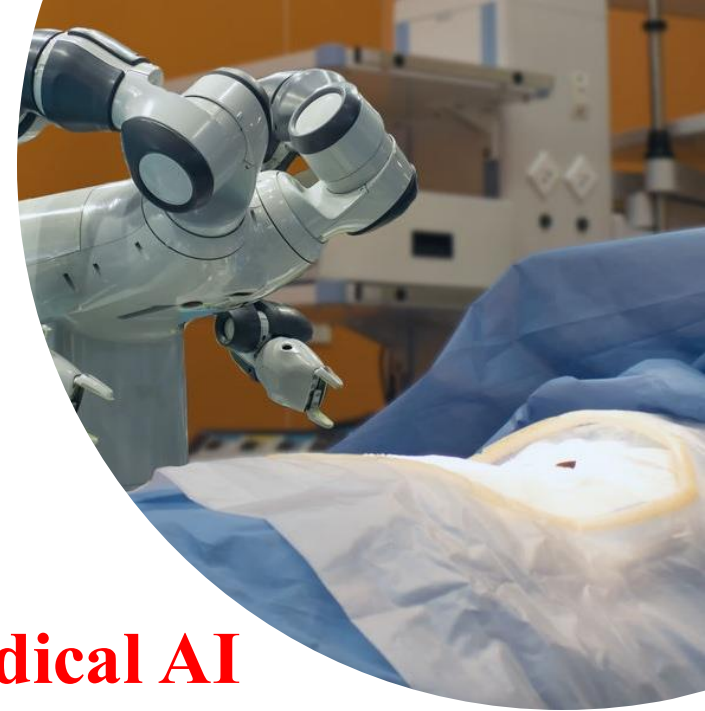


该二维码7天内(9月8日前)有效, 重新进入将更新

Self-driving



AI Teaching

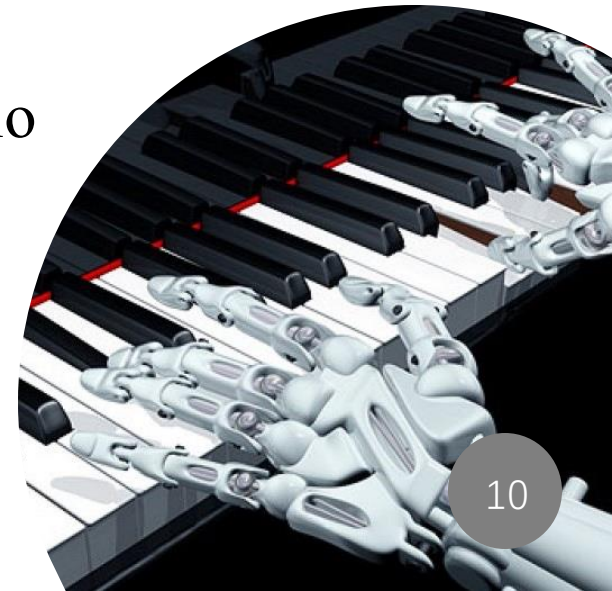


Medical AI

AI for Computer Vision



AI for Audio



AI changes the world!

Embodied AI



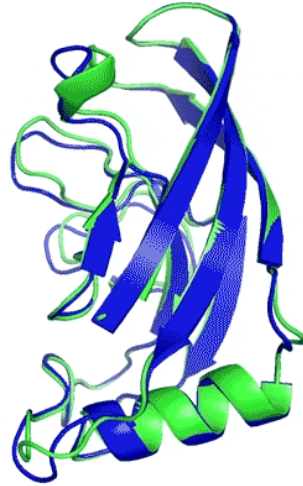
Automatic Driving



AI for Drug Design



T1037 / 6vr4
90.7 GDT
(RNA polymerase domain)



T1049 / 6y4f
93.3 GDT
(adhesin tip)

- Experimental result
- Computational prediction

Results of AlphaFold



LLM participates in China's college entrance exam (Gaokao):

新课标卷得分情况											
模型	语文	数学	英语	物理	化学	生物	历史	地理	思想政治	理科总分	文科总分
InternLM2-20B-WQX	112	74	138.5	39	48	57	82	58	67	468.5	531.5
GPT-4o	111.5	73	141.5	36	40	65	88	59	58	467	531
Qwen2-72B	124	68	139	42	44	48	85	70	60	465	546
Yi-1.5-34B	97	31	134.5	21	37	49	48	41	51	369.5	402.5
Qwen2-57B	99.5	58	126.5	7	6	51	73	4	62	348	423
GLM-4-9B	86	48	97	18	27	67	80	62	48	343	421
Mixtral 8x22B	77.5	21	116.5	25	35	46	54	56	38	321	363

AIGC: 迪迦渣渣辉



Imagine your team has 10 billion RMB, what AI will you build?

- Discuss with a group
- Fill in a short answer by scanning (扫码回答)

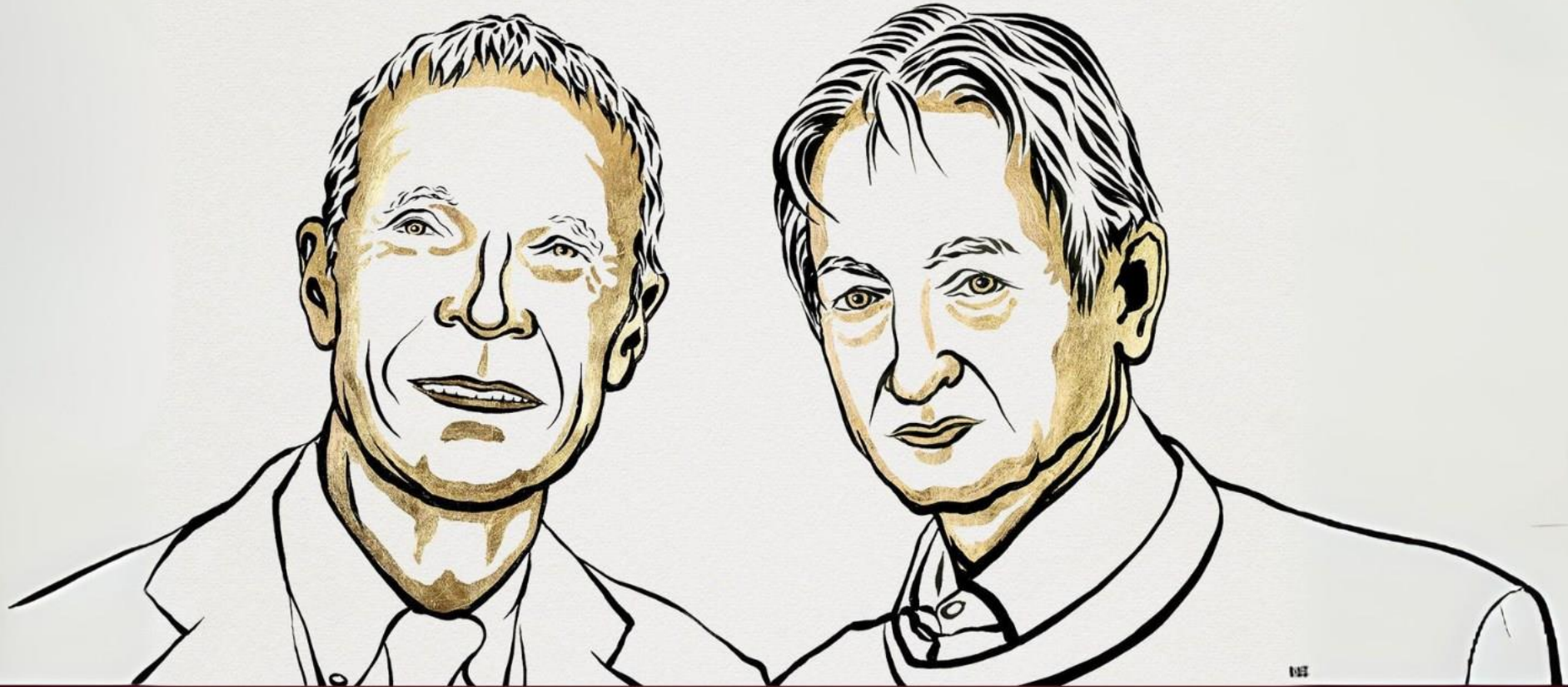


Outline of this week

1. Info about This Course

2. Introduction to Artificial Intelligence (AI)

THE NOBEL PRIZE IN PHYSICS 2024



John J. Hopfield

Geoffrey E. Hinton

“for foundational discoveries and inventions
that enable machine learning
with artificial neural networks”

THE ROYAL SWEDISH ACADEMY OF SCIENCES

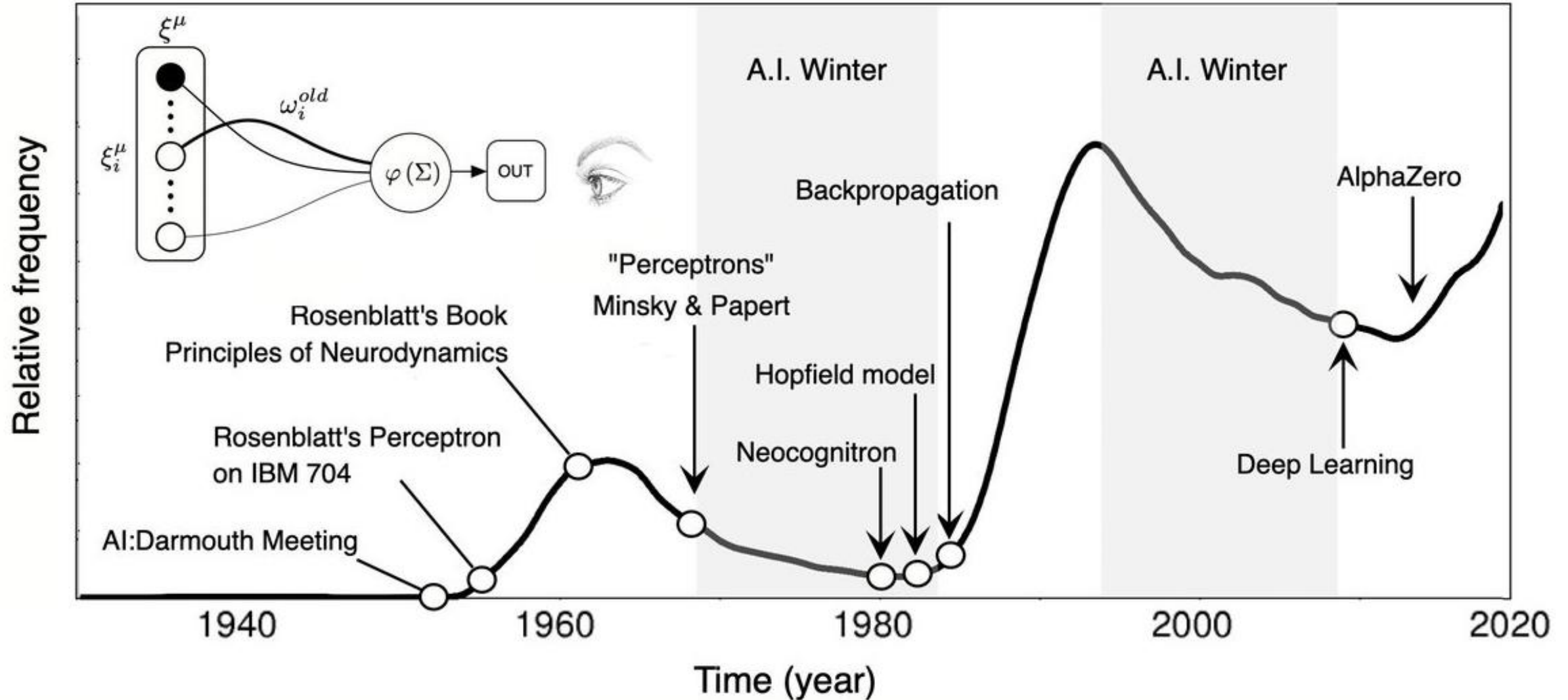
Please guess when artificial neural networks are firstly proposed?

- A. 1940s
- B. 1960s
- C. 1980s
- D. 2000s



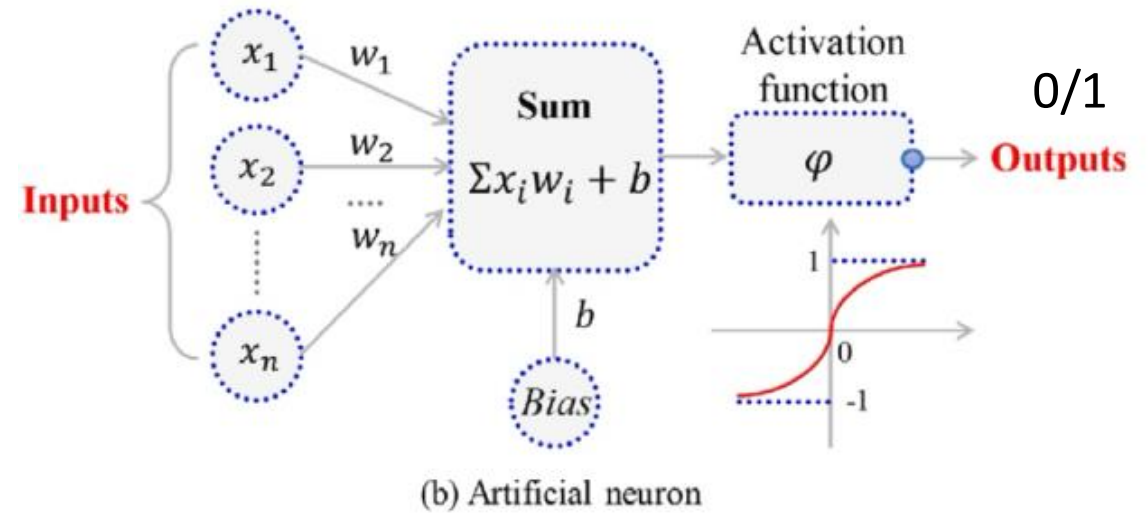
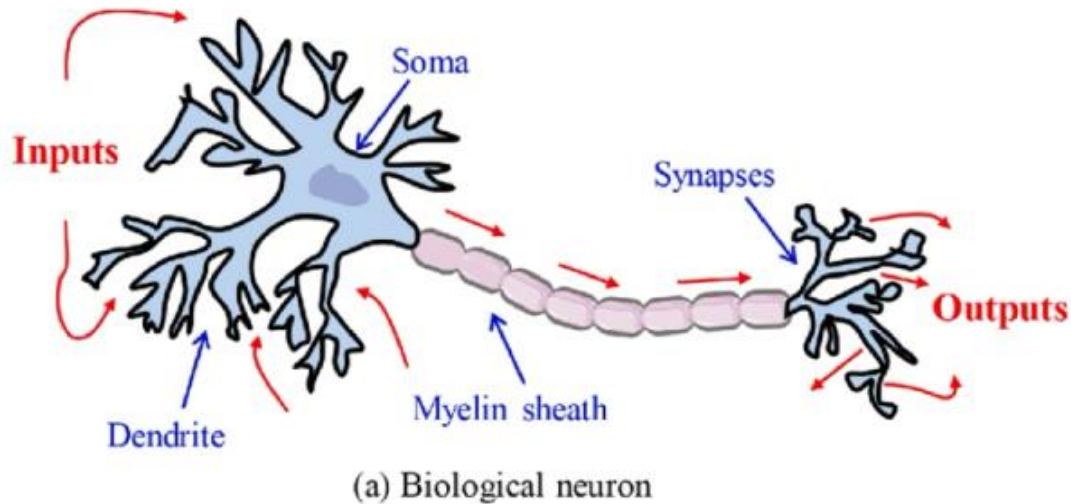
Scan to answer
(扫码回答)

Development of artificial Neural Networks



Perceptron

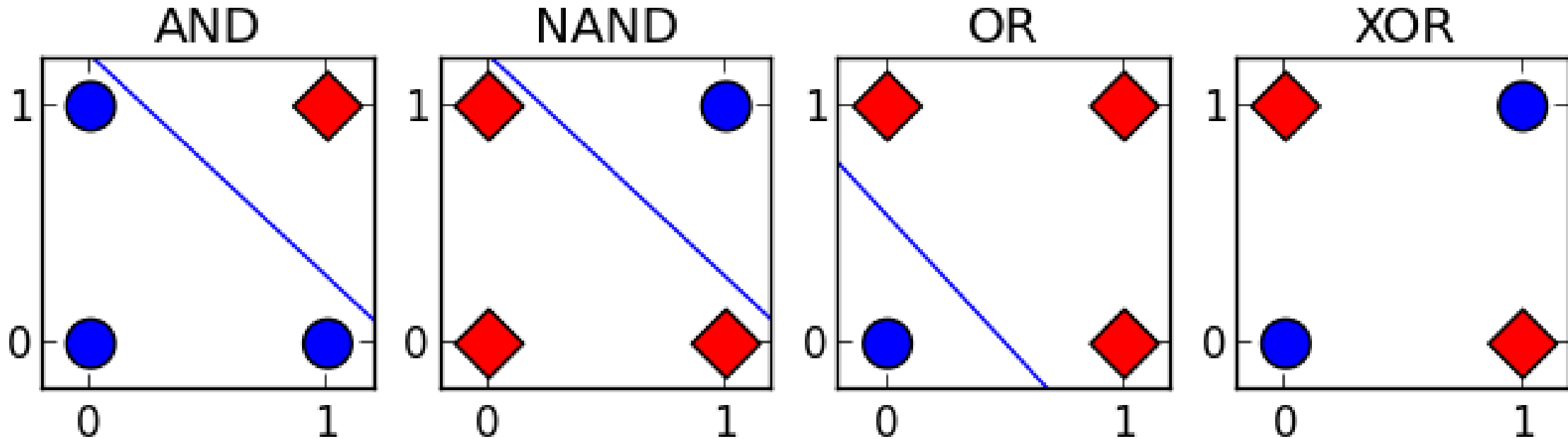
Inspired by the human neural network, the perceptron is developed from 1943 to 1958. Perceptron triggered widespread enthusiasm about the potential of ANN.



1st ANN Winter

Marvin Minsky & Seymour Papert published "**Perceptrons**" (1969):

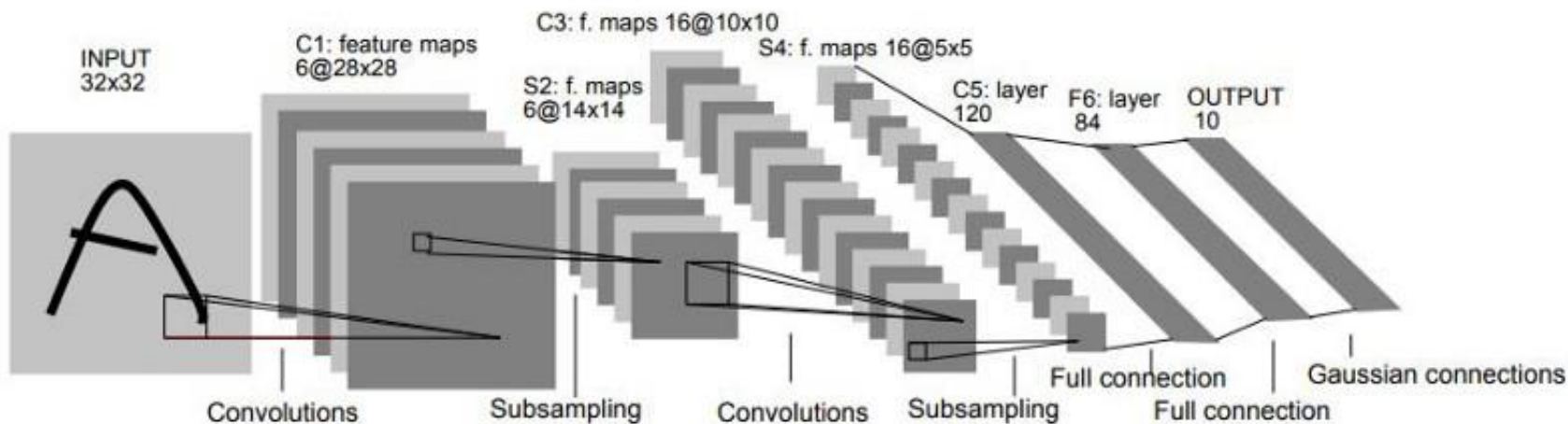
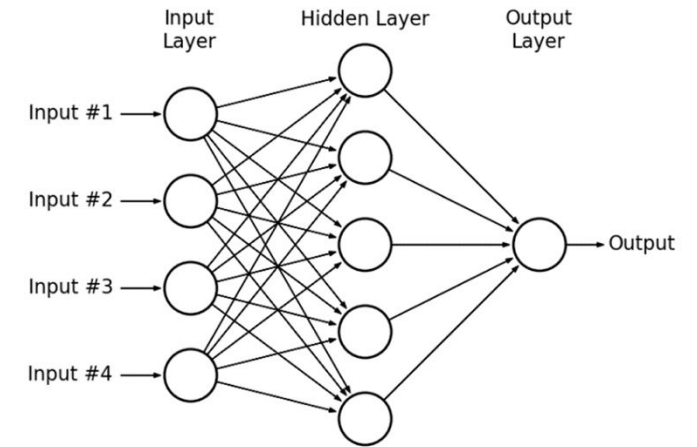
- Single-layer perceptron cannot solve non-linear problems (XOR)
- Led to funding cuts and decline in ANN research.



Revival with Multi-Layer Perception

1980s-1990s:

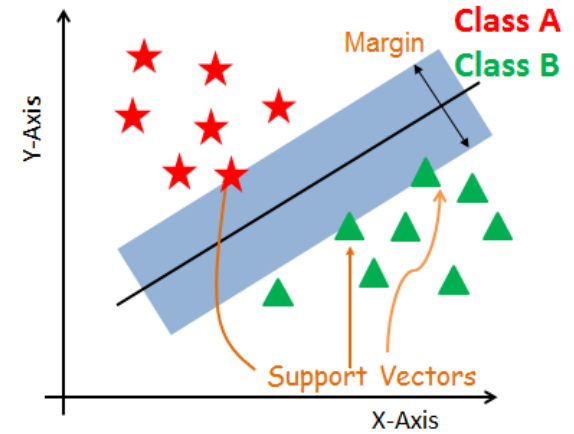
- Add a hidden layer to solve the XOR problem
- Hinton proposed back propagation to optimize MLP
- Yang LeCun develop handwritten digit recognition for ATM



2nd ANN Winter

Late 1990s

- The rise of support vector machine (SVM)
- The limitation of computational resource



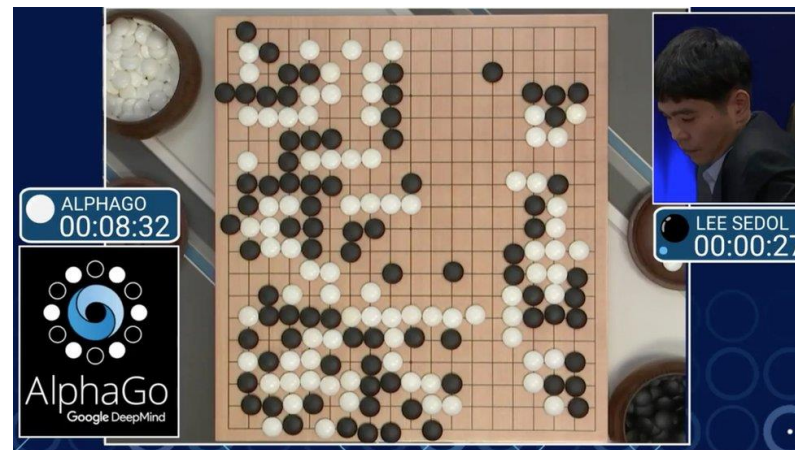
Deep Learning

2006: Layerwise pre-training of deep networks (Hinton et al.)

2012: AlexNet obtains huge gains in imageNet



2016: AlphaGo uses deep reinforcement learning, defeat world champion Lee Sedol in Go



YOSHUA BENGIO,
GEOFFREY E. HINTON
AND YANN LECUN

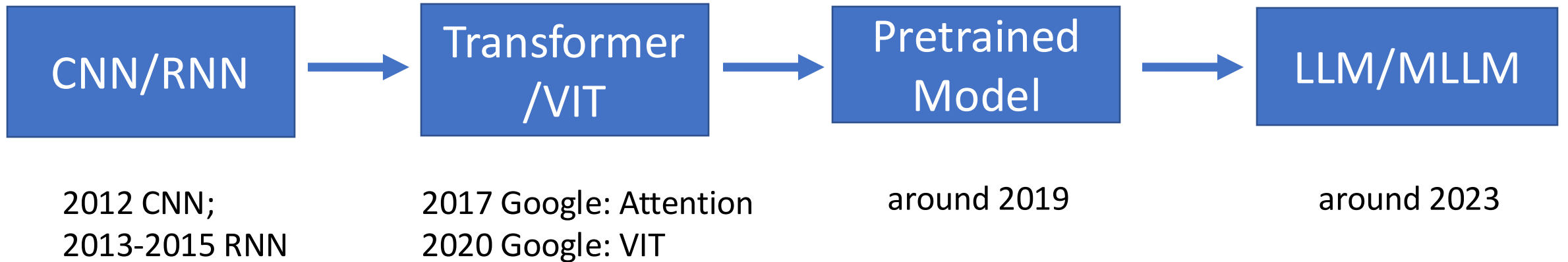
For conceptual and engineering
breakthroughs that have made
deep neural networks a critical
component of computing

A.M.
TURING

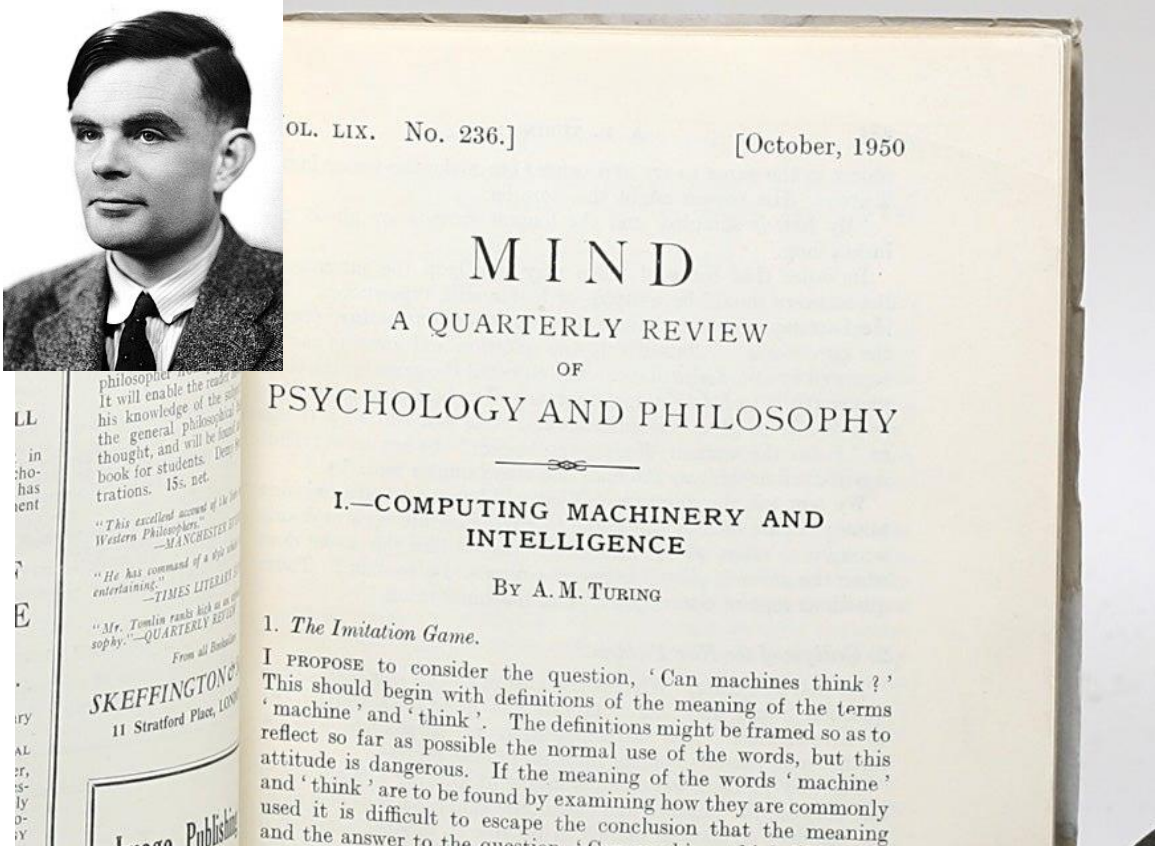
A W A R D
2018



Recent Development History



Turing Test



Alan Turing (1912 – 1954)

- Turing asked : Can machines think?
- Propose Turing Test

How Turing Test Works?

*"Can machines think?" → Instead of debating philosophy, Turing said:
"If a machine acts indistinguishably from a human, we should consider it intelligent."*

1. Setup:

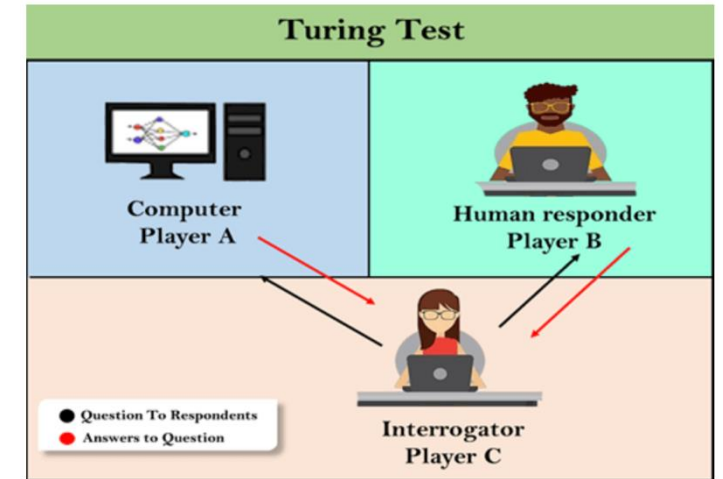
1. A **human interrogator** (审讯员) converses via text with two hidden entities:

- One **human**
- One **machine** (AI)

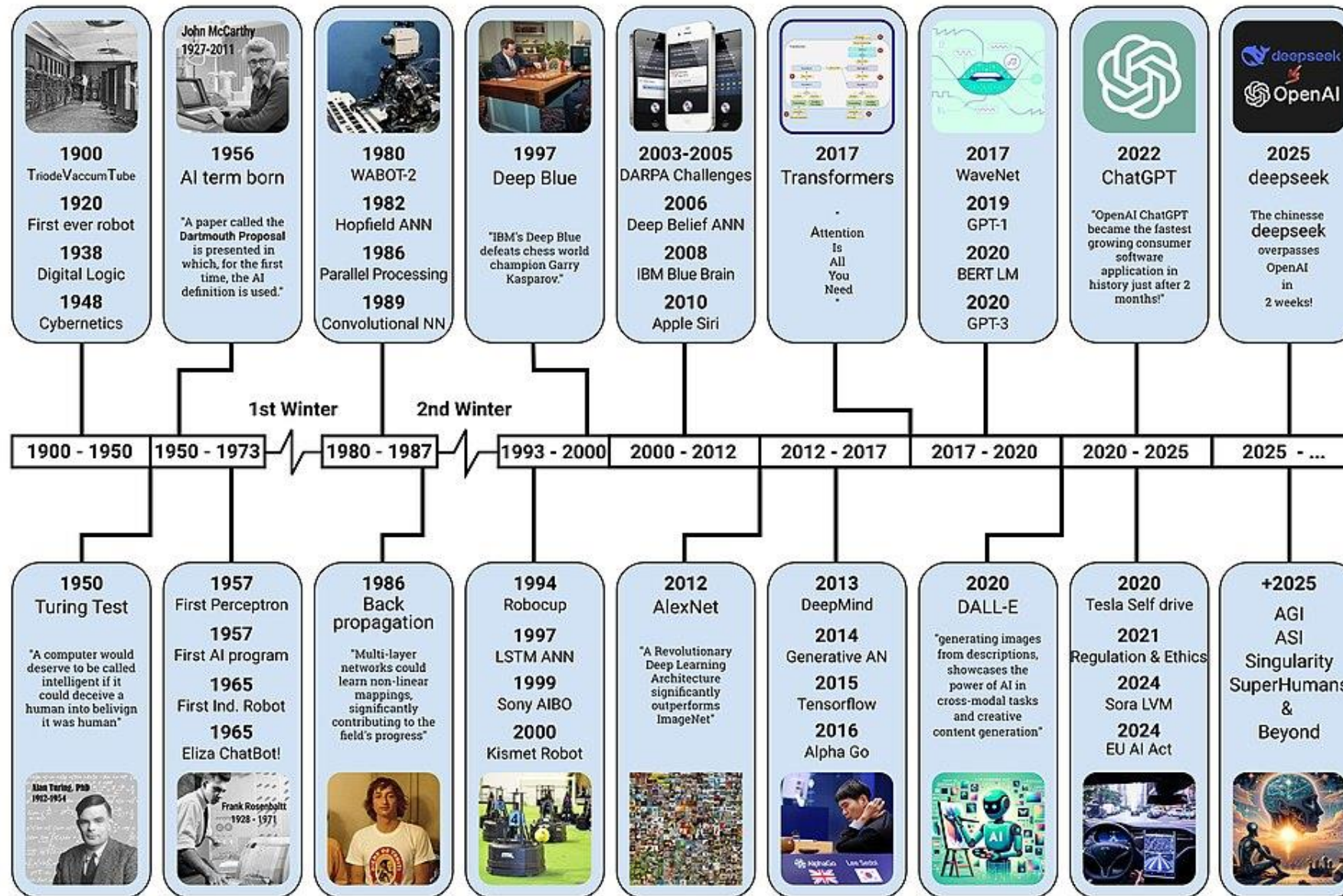
1. All communication is text-only (no voice or visual cues).

2. Goal:

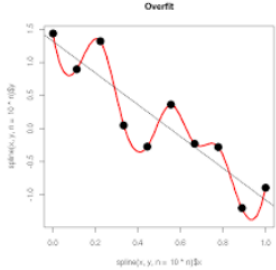
1. The interrogator asks questions to both entities.
2. If the interrogator **cannot reliably distinguish** the machine from the human (e.g., <50% accuracy after 5 mins), the machine **passes the test**.



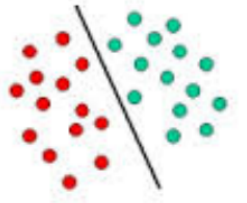
Summary of AI History



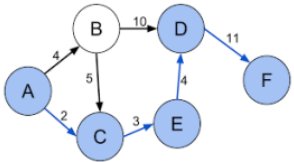
Early Ideas about AI



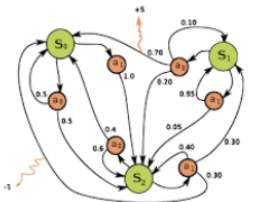
1801: linear regression (Gauss, Legendre)



1936: linear classification (Fisher)



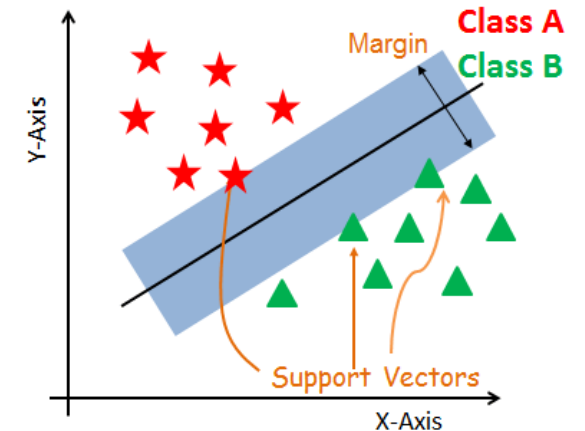
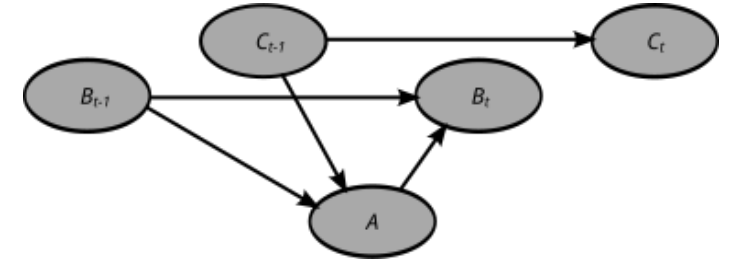
1956: Uniform cost search for shortest paths (Dijkstra)



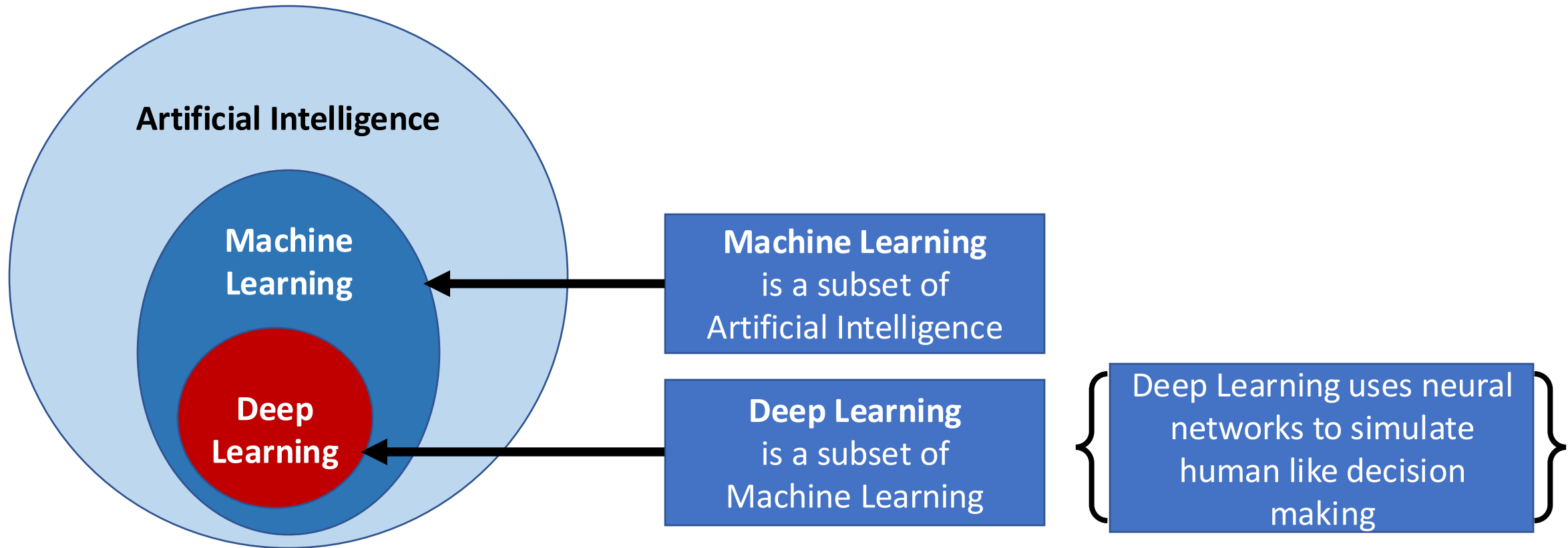
1957: Markov decision processes (Bellman)

Statistical Machine Learning

- 1985: Bayesian networks (Pearl)
- 1995: Support vector machine (Cortes/Vapnik)

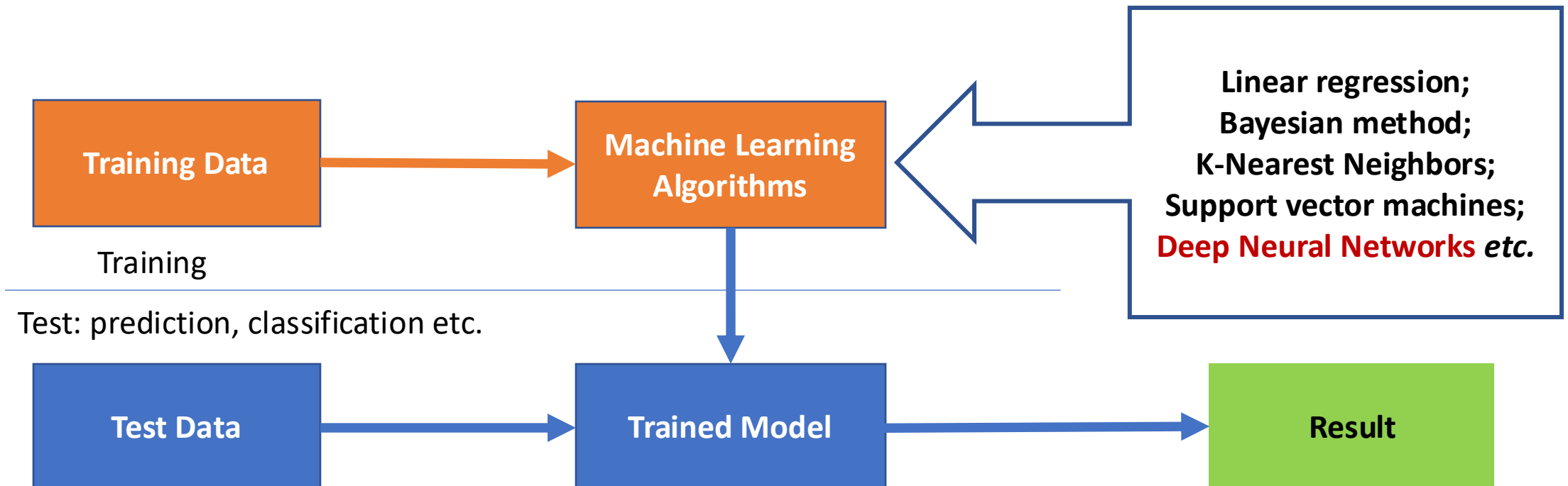


Subsets of Artificial Intelligence

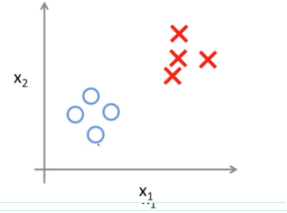


What is Machine Learning?

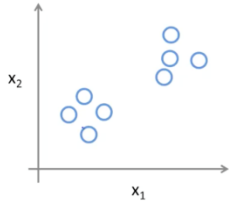
- Type of Artificial Intelligence that provides computers with the ability to **learn without being explicitly programmed**.
- **Various methods** can be used to make predictions based on data.



Machine Learning Approaches ?



- **Supervised Learning:** Learning with a **labelled** training set
 - Example 1: email spam detector with training set of labelled emails
 - Example 2: classify the benign tumor or cancer given the labels that tell the model this is a benign tumor or cancer in the training stage



- **Unsupervised Learning:** Discovering patterns in **unlabelled** dataset
 - Example: cluster similar diseases based on the Medical Images
- **Semi-supervised Learning: Between** Supervised and Unsupervised Learning
- **Reinforcement Learning:** Learning based on **feedback** or reward
 - Example: learn to play chess by winning or losing

